CSA 0378-DATA STRUCTURES

24. KRUSHKAL'S ALGORITHM

main.c   Share   Run

```c
#include <stdio.h>

struct Edge {
    int u, v, w;
};

int parent[10];

int find(int i) {
    while(parent[i] != i)
        i = parent[i];
    return i;
}

void unionSet(int a, int b) {
    parent[a] = b;
}

int main() {
    struct Edge edge[5] = {
        {0, 1, 2},
        {0, 2, 3},
        {1, 2, 1},
        {1, 3, 4},
        {2, 3, 5}
    };

    int i, count = 0, cost = 0;
    int min, a, b, u, v;

    for(i = 0; i < 4; i++)
        parent[i] = i;

    printf("Edges in MST:\n");

    while(count < 3) {
        min = 999;

        for(i = 0; i < 5; i++) {
            if(edge[i].w < min) {
                min = edge[i].w;
                a = edge[i].u;
                b = edge[i].v;
            }
        }

        u = find(a);
        v = find(b);

        if(u != v) {
            printf("%d - %d : %d\n", a, b, min);
            cost += min;
            count++;
            unionSet(u, v);
        }

        for(i = 0; i < 5; i++) {
            if(edge[i].u == a && edge[i].v == b && edge[i].w == min) {
                edge[i].w = 999;
                break;
            }
        }
    }

    printf("Total Cost = %d\n", cost);

    return 0;
}
```

Output

```
Selected Edges:
B-C = 1
A-B = 2
B-D = 4
Total Cost = 7

=== Code Execution Successful ===
```